Aryan Gupta

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EDUCATION

Birla Institute of Technology, Mesra — **2023 – 2027**
B.Tech in Electronics and Communication Engineering — CGPA: 8.49/10

Angels High School, Hazaribagh, *Class XII, CBSE* — 2022 | 88.67%
Angels High School, Hazaribagh, *Class X, CBSE* — 2020 | 91.67%

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL
Backend & Full-Stack: Node.js, Express.js, FastAPI, React.js, REST APIs, JWT Authentication, SQLAlchemy, Mongoose, TanStack React Query
Databases: PostgreSQL, MySQL, MongoDB, Redis
AI & ML: Generative AI, Machine Learning, Deep Learning, Sentence Transformers, LLMs, RAG Pipelines, LangChain, Vector Search (FAISS)
Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions (CI/CD), PM2, Vercel, Render
Developer Tools: Git, GitHub, Linux, Postman
Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

OPEN SOURCE CONTRIBUTIONS

w3lib (Scrapy ecosystem) | *Python, HTML parsing* — **Merged** | **PR #379**

- Fixed HTML closing-tag parsing in Scrapy's **w3lib**, correcting **base-URL** and **meta-refresh** extraction; added **regression tests** for malformed HTML and string/byte inputs.

node-cron (Node.js task scheduler) | *TypeScript, Node.js* — **Merged** | **PR #622**

- Fixed a scheduler bug where failed jobs bypassed **maxExecutions** and continued running; enforced execution limits in a **finally** block and added **regression tests** for thrown errors and rejected promises.

EXPERIENCE

QOE Technologies and Services — **May 2026 – June 2026**
Software Developer Intern — *Remote*

- Developed **full-stack** booking features for Soul Consciousness Lab using **React.js**, **Node.js**, **Express.js**, and **MongoDB**; reduced duplicate booking records by **90%** in controlled repeated-click tests through submission locking and backend duplicate checks.
- Reduced page load time by **37% (3.8s to 2.4s)** through **lazy loading**, removal of redundant API requests, and optimized **React** re-renders.

PROJECTS

DocuMind – RAG Document Assistant | *Python, FastAPI, PostgreSQL, Redis, FAISS, LangChain, React, Docker* — **GitHub** | **Demo**

- Built an end-to-end **RAG pipeline** over PDF, DOCX, and TXT files, using **SentenceTransformer** embeddings and a **FAISS** vector store for top-k semantic retrieval.
- Developed a modular **FastAPI** REST backend with **Pydantic**, **SQLAlchemy**, and **PostgreSQL** for document metadata, query history, and analytics; containerized with **Docker**.
- Implemented multi-turn **LLM** Q&A with session-aware prompts, **smart follow-up suggestions**, and real-time streaming to a **React** client; added **Redis** rate limiting and query caching (1-hour TTL), cutting latency from **2.5s to under 15ms**.

DPI Engine – Deep Packet Inspection System | *C++, PCAP, Multithreading, TLS SNI, Flow Tracking* — **Group Project** | **GitHub**

- Co-engineered a **multithreaded C++17** deep packet inspection (DPI) engine from scratch, with a custom **PCAP** reader and Ethernet/IPv4/TCP/UDP/ICMP parsers; thread-safe queues sustain **10,000+ packets/sec** on a single core.
- Co-built traffic classification that extracts **SNI** from TLS 1.2/1.3 Client Hello packets and decodes **HTTP Host** headers and **DNS** queries to map domains to applications.
- Co-implemented a bidirectional **five-tuple flow tracker** and a **rule engine** that blocks traffic by IP, wildcard domain, application, and port, emitting a filtered PCAP.

Tweetie – Social Media Platform | *React.js, Node.js, Express.js, MongoDB, REST APIs, JWT, AWS S3* — **GitHub** | **Live Demo**

- Engineered a **full-stack** social media platform with stateless **JWT** authentication in **HTTP-only cookies** to mitigate **XSS** token theft, using **AWS S3** for scalable media storage and **TanStack React Query** caching to cut redundant network requests.
- Built a custom **time-decay content-ranking algorithm** to personalize the user feed, with an asynchronous **node-cron** worker that precomputes rankings for **O(1)** feed read latency; deployed on **AWS EC2** (Ubuntu) with **PM2** for zero-downtime availability.

ACHIEVEMENTS

- Solved **1000+ DSA problems** across LeetCode, Codeforces, and CodeChef.
- Achieved **Knight** on LeetCode (**Max Rating 1943**, Top 3.5%), **Expert** on Codeforces (1657), and **3-Star** on CodeChef (Max Rating 1639).
- **Global Rank 285** in LeetCode Weekly Contest 502 and **1,488** in Biweekly Contest 184.
- Qualified for Round 2 of **Google The Big Code 2026** (Top 1,500).
- **Captain**, BIT Mesra Football Team (2025 – Present), demonstrating leadership and teamwork.